



Statistics of Financial Markets

Chapter 13: Machine learning



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Acronyms used in this material (1/2)

- ▣ **AI:** Artificial Intelligence
- ▣ **ARCH:** AutoRegressive Conditional Heteroskedasticity
- ▣ **AUC:** Area Under the (ROC) Curve
- ▣ **BET:** Bucharest Exchange Trading (index)
- ▣ **BVB:** Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ **CV:** Cross-Validation
- ▣ **DM:** Diebold–Mariano (test)
- ▣ **DSR:** Deflated Sharpe Ratio
- ▣ **EODHD:** EOD Historical Data (market-data provider)
- ▣ **GARCH:** Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ **GB:** Gradient Boosting
- ▣ **GJR-GARCH:** Glosten–Jagannathan–Runkle GARCH
- ▣ **HAR:** Heterogeneous AutoRegressive (model)
- ▣ **HS:** Historical Simulation
- ▣ **IBM:** International Business Machines (company)
- ▣ **MA:** Moving Average
- ▣ **ML:** Machine Learning
- ▣ **MLP:** MultiLayer Perceptron
- ▣ **MSE:** Mean Squared Error
- ▣ **NN:** Neural Network (NN k : k hidden layers)

Acronyms used in this material (2/2)

- ▣ **NN-ARCH**: Neural-Network ARCH
- ▣ **NN1**: Neural Network with 1 hidden layer
- ▣ **NN3**: Neural Network with 3 hidden layers
- ▣ **NN4**: Neural Network with 4 hidden layers
- ▣ **OLS**: Ordinary Least Squares
- ▣ **OLS-3**: OLS with 3 predictors (size, book-to-market, momentum)
- ▣ **OMV**: Österreichische Mineralölverwaltung — Austrian Mineral Oil Administration (OMV group)
- ▣ **PSR**: Probabilistic Sharpe Ratio
- ▣ **QGB**: Quantile Gradient Boosting
- ▣ **QLIKE**: Quasi-LIKElihood (loss function)
- ▣ **QR**: Quantile Regression
- ▣ **RBF**: Radial Basis Function (network)
- ▣ **RF**: Random Forest
- ▣ **RMSE**: Root Mean Squared Error
- ▣ **ROC**: Receiver Operating Characteristic (curve)
- ▣ **RV**: Realised Variance
- ▣ **SFE**: Statistics of Financial Markets, Exercises (Quantlet collection of the textbook)
- ▣ **SHAP**: SHapley Additive exPlanations
- ▣ **SSE**: Sum of Squared Errors
- ▣ **US**: United States

Today's Question and Route

- ▣ **Question:** can flexible prediction methods forecast financial risk and returns better than the models of Chapters 7–10, and how do we check it honestly?
 - ▶ **machine learning** (ML): algorithms that learn a prediction rule from data, judged by their accuracy on new data
- ▣ **Route** of the chapter
 - ▶ prediction against inference; overfitting, bias and variance; validation for time series without look-ahead bias
 - ▶ ridge and lasso; trees, random forest, gradient boosting; neural networks and the neural-network ARCH of the textbook
 - ▶ three applications: realised volatility, the sign of returns, VaR 1%; data snooping; a landmark paper

Learning Outcomes

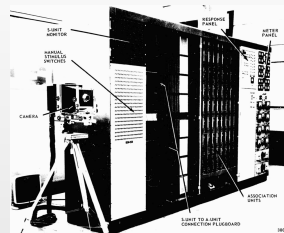
- Explain the difference between prediction and inference, and the bias–variance trade-off
- Design a walk-forward evaluation with purging, and recognise look-ahead bias and leakage
- Fit ridge, lasso, a regression tree, a random forest, gradient boosting and a small neural network
- Compare forecasts with out-of-sample R^2 , QLIKE and the Diebold–Mariano test, against a proper baseline
- Forecast VaR 1% by quantile regression and correct a Sharpe ratio for the number of strategies tried

Reading and Tools

- Textbook: [Franke, Härdle & Hafner \(2019\)](#), Ch. 19 (neural networks)
 - ▶ statistical learning in Python: [James et al. \(2023\)](#); in depth: [Hastie, Tibshirani & Friedman \(2009\)](#)
 - ▶ machine learning in finance: [Kelly & Xiu \(2023\)](#); [López de Prado \(2018\)](#)
- Python Quantlets of this chapter: [Quantlets/Ch_13](#)
 - ▶ ported from the SFE Quantlets [SFEnnarch](#) and [SFEnnpyusd](#); models from scikit-learn (small, fixed seeds, a few minutes in Colab)
- Lecture notebook: [open in Google Colab](#)
- Video courses: [!\[\]\(5ba1bc70d78f05c00988641e5e513c62_img.jpg\) Machine learning in Financial Risk](#); [!\[\]\(0d3dd579ab24f8020cd6c2659f3acb8c_img.jpg\) Random Forests](#)

The Perceptron, 1958

- **Rosenblatt (1958)**: a machine that learns to classify images by adjusting weights after each error
 - ▶ the **perceptron**: output 1 if $w_0 + w^T x > 0$, else 0
 - ▶ x : the inputs (photocell signals); w : their weights, learned from examples; w_0 : the threshold term; $w^T x = \sum_j w_j x_j$
 - ▶ the Mark I Perceptron (Cornell, 1960): 400 photocells, weights set by electric motors
- Three later milestones
 - ▶ 1986: back-propagation trains networks with hidden layers [Rumelhart, Hinton & Williams \(1986\)](#)
 - ▶ 2001: random forests [Breiman \(2001b\)](#) and gradient boosting [Friedman \(2001\)](#)
 - ▶ 2024: Nobel Prize in Physics to Hopfield and Hinton for neural networks



The Mark I Perceptron, from its operator's manual

Figure: J.C. Hay, A.E. Murray (1960); public domain;
[Wikimedia Commons](#)

Two Cultures of Statistical Modelling

- Breiman (2001a), a statistician at Berkeley, describes two cultures
 - ▶ **data modelling**: assume a stochastic model (the linear regression $y = x^\top \beta + \varepsilon$, GARCH), estimate it, test its parameters
 - ▶ **algorithmic modelling**: treat the mechanism as unknown; find a rule $\hat{f}$ that predicts y well on new data
- This course used the first culture in [Chapters 2–11](#)
 - ▶ this chapter adds the second, and asks when it pays off in finance



Evans Hall, University of California, Berkeley,
home of the Department of Statistics

Photo: [Gabriel Classon \(2021\)](#); CC BY 2.0; [Wikimedia Commons](#)

Prediction against Inference

- **Inference:** is $\beta_j \neq 0$? how large is the effect? (standard errors, tests, [Chapter 6](#))
 - ▶ example: does the leverage effect exist (GJR-GARCH, [Chapter 9](#))?
- **Prediction:** how close is $\hat{y}$ to y on data the model has never seen?
 - ▶ example: next week's volatility; tomorrow's VaR 1%
 - ▶ the coefficients may be uninterpretable; what matters is the error on new data
- The two goals can conflict
 - ▶ a biased estimator (ridge, lasso) can predict better than the unbiased OLS
 - ▶ a model that predicts well does not prove a causal mechanism

Supervised Learning: Notation and Loss (1/2)

- Data (x_t, y_t) , $t = 1, \dots, n$
 - ▶ t : the index of the observation (a day, a week); n : the number of observations
 - ▶ $x_t \in \mathbb{R}^p$: the **features**, p numbers known at time t (past volatilities, past returns)
 - ▶ y_t : the **target**, the quantity we want to forecast (next week's volatility)
- Two types of problems
 - ▶ **regression**: y continuous (volatility, return)
 - ▶ **classification**: y a class (up/down, default; [Chapter 12](#))

Supervised Learning: Notation and Loss (2/2)

- Learning = choose $\hat{f}$ in a family $\mathcal{F}$ that minimises the average **loss** $\frac{1}{n} \sum_t L(y_t, f(x_t))$
 - ▶ $f(x_t)$: the forecast of a rule f for observation t ; $L(y, \hat{y})$: the cost of forecasting $\hat{y}$ when y is observed
 - ▶ $\mathcal{F}$: the candidate rules (all linear regressions, all trees of depth 3); $\hat{f}$: the rule chosen from the data
- Losses used in this course
 - ▶ squared error $(y - \hat{y})^2$ (MSE: mean squared error); QLIKE for variances ([Chapter 8](#))
 - ▶ quantile (pinball) loss for VaR ([Chapter 10](#)); log loss for probabilities ([Chapter 12](#))
- A **hyperparameter** is chosen before the fit
 - ▶ examples: tree depth, penalty λ , number of neurons
 - ▶ the parameters (coefficients, weights) are learned by the fit

Where ML Helps in Finance, and Where It Struggles

- ▣ **Helps:** targets with structure and enough data
 - ▶ volatility is persistent ([Chapters 8–9](#)): a lot to predict, nonlinear effects possible
 - ▶ cross-sections with thousands of stocks and many characteristics (the case study)
- ▣ **Struggles:** a low **signal-to-noise ratio**
 - ▶ daily returns are close to unpredictable ([Chapter 7](#)): a flexible model fits the noise
 - ▶ markets change (regimes, crises): the past is an imperfect training set
 - ▶ one price history: we cannot repeat the experiment, and the observations are dependent

Recap: What Machine Learning Adds

- ▣ ML judges a model by its error on new data, not by the significance of its coefficients
- ▣ Features x_t , target y_t , loss L , hyperparameters chosen before the fit
- ▣ Finance: easy to predict volatility, hard to predict returns

Training Error and Test Error

- **Training error:** the average loss on the data used for the fit; **test error:** on new data from the same process
 - ▶ the training error always falls as the model becomes more flexible
- **Overfitting:** the model learns the noise of the training sample
 - ▶ symptom: a small training error and a large test error
 - ▶ a tree with one leaf per observation has zero training error and predicts badly
- **Underfitting:** the model is too rigid to capture the signal
 - ▶ example: a constant forecast of volatility in a world with volatility clustering

The Bias–Variance Decomposition (1/2)

- Model: $y = f(x) + \varepsilon$, $E[\varepsilon] = 0$, $\text{Var}(\varepsilon) = \sigma^2$
 - ▶ f : the true, unknown relation between x and y ; ε : the noise, unpredictable from x , with variance σ^2
 - ▶ $\hat{f}$: the estimate of f from a random training sample; another sample would give another $\hat{f}$
- Expected test error at a new point x_0 , with $y_0 = f(x_0) + \varepsilon_0$:

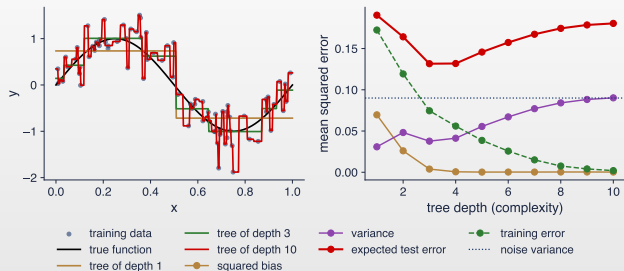
$$E[(y_0 - \hat{f}(x_0))^2] = \underbrace{(E[\hat{f}(x_0)] - f(x_0))^2}_{\text{bias}^2} + \underbrace{\text{Var}(\hat{f}(x_0))}_{\text{variance}} + \underbrace{\sigma^2}_{\text{noise}}$$

- ▶ $E[\cdot]$: the average over all possible training samples and over the noise ε_0
- ▶ bias: how far the average estimate is from the truth; variance: how much $\hat{f}(x_0)$ moves from one sample to another

The Bias–Variance Decomposition (2/2)

- ▣ Derivation: write $y_0 - \hat{f} = (f - E\hat{f}) + (E\hat{f} - \hat{f}) + \varepsilon_0$, all at x_0
 - ▶ square and take the expectation: the three squares give bias^2 , variance and σ^2
 - ▶ the cross terms have mean 0: $E[E\hat{f} - \hat{f}] = 0$, and ε_0 is independent of the training sample
- ▣ Reading the decomposition
 - ▶ a rigid model: large bias, small variance
 - ▶ a flexible model: small bias, large variance
 - ▶ σ^2 is the floor: no model goes below it; in daily returns σ^2 is almost everything

Bias and Variance of Regression Trees



- Simulation: $y = \sin(2\pi x) + \varepsilon$, $\sigma^2 = 0.09$, $n = 80$, 300 training samples; left: one sample with trees of depth 1, 3 and 10
- Depth 1: bias² 0.070, variance 0.031; depth 10: bias² 0.000, variance 0.090; the test error is smallest at depth 3 (0.132)

Interpretation: the U-Shaped Test Error

- The training error falls from 0.172 to 0.002; the test error falls, then rises
 - ▶ depth 10 has almost zero training error and a test error of 0.180, worse than depth 3
- Consequences for practice
 - ▶ never select a model by its training error
 - ▶ complexity is a hyperparameter: choose it on data that were not used for the fit
 - ▶ in finance the noise is large, so the best model is usually simpler than in this simulation

Recap: Overfitting, Bias and Variance

- Test error = $\text{bias}^2 + \text{variance} + \text{noise}$
- More flexibility: less bias, more variance; the training error is always too optimistic
- The amount of flexibility must be chosen on new data

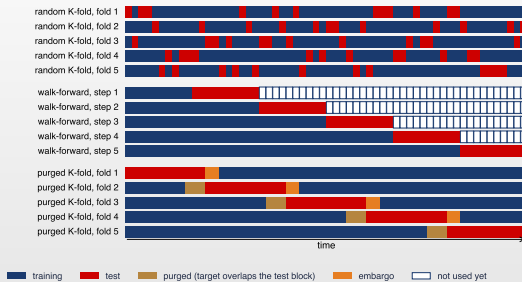
Training, Validation and Test Samples

- Three roles for the data
 - ▶ **training**: estimate the parameters of each candidate model
 - ▶ **validation**: choose the hyperparameters and the model
 - ▶ **test**: measure the error of the final choice, once
- Every look at the test sample turns it into a validation sample
 - ▶ trying ten models on the test data and reporting the best is model selection, not testing
- **K-fold cross-validation**: split the data into K folds; each fold is the validation set once, the others train the model
 - ▶ valid for i.i.d. data; for a time series only under strong conditions (no dependence left in the errors) [Bergmeir, Hyndman & Koo \(2018\)](#)

Look-Ahead Bias and Leakage

- **Look-ahead bias:** the forecast for day t uses information that was not available at the close of day $t - 1$
 - ▶ features computed with future prices (a centred moving average, a full-sample mean)
 - ▶ scaling or standardising with statistics of the whole sample; revised data; today's index members (survivorship)
- **Leakage:** information about the test target reaches the training sample
 - ▶ overlapping targets: the 21-day return from day t and from day $t + 1$ share 20 days
 - ▶ a random fold puts day $t + 1$ in training and day t in test: the model has seen the answer
- Both make a model look better than it can be in real time

Three Validation Schemes

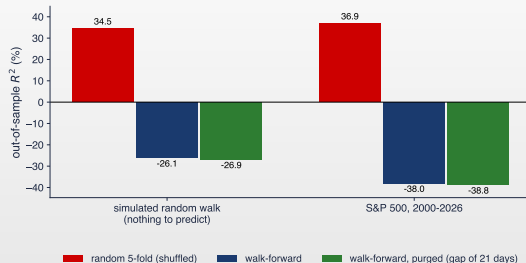


- **Walk-forward:** train on the past, test on the next block, move forward; the expanding window keeps all the past
- **Purged K-fold with embargo** López de Prado (2018): drop the training rows whose targets overlap the test block (purging) and a few rows after it (embargo)

Walk-Forward Validation, Step by Step

- The design used in this chapter for the S&P 500 (data from 8 April 2008)
 - ▶ one fold per calendar year from 2013: 14 folds; the model is re-estimated each January
 - ▶ training window: all earlier days (expanding); test: every day of the year
- Purging: the target is the mean variance of the next 5 days
 - ▶ for 2013 the last 5 training rows (from 24 December 2012) are dropped: their targets reach into 2013
 - ▶ the 2013 fold trains on 1 188 days and tests on 252; the 2026 fold trains on 4 458 days
- Inside each training window, hyperparameters are tuned by a walk-forward split of the training data only
 - ▶ no embargo is needed: the test block always comes after the training block

Leakage in Practice: Random Folds on Overlapping Targets



- Random forest for the next 21-day return from past returns and volatilities; out-of-sample R^2 against the training mean
- Simulated random walk: random 5-fold 34.5%, walk-forward -26.1%; S&P 500 (6 631 days): 36.9% and -38.0%

Interpretation: a Forecast of Pure Noise with $R^2 > 30\%$

- The random walk has nothing to predict, yet random folds report $R^2 = 34.5\%$
 - ▶ the forest memorises neighbouring days, whose targets share 20 of 21 returns with the test day
- Walk-forward gives a negative R^2 : the forest fits noise and loses against the training mean
 - ▶ purging (a gap of 21 days) changes little here, because in walk-forward only the boundary rows overlap
- Rule: with overlapping targets or dependent data, never shuffle the time order

Recap: Validation for Time Series

- ▣ Training, validation and test have different jobs; the test sample is used once
- ▣ Walk-forward with purging respects the arrow of time; random K-fold leaks with overlapping targets
- ▣ Look-ahead bias and leakage produce impressive but useless results

Many Features, Unstable OLS

- OLS (ordinary least squares): $\hat{\beta} = \arg \min_{\beta} \sum_t (y_t - x_t^{\top} \beta)^2 = (X^{\top} X)^{-1} X^{\top} y$
 - ▶ X : the $n \times p$ matrix whose row t is $x_t^{\top}$; y : the vector of targets; $\arg \min_{\beta}$: the β that minimises the sum
 - ▶ unbiased, but its variance $\sigma^2 (X^{\top} X)^{-1}$ (σ^2 : the error variance) explodes when features are many or strongly correlated
- Volatility features are highly correlated: today's, this week's and this month's realised variance
 - ▶ OLS may give large coefficients of opposite signs that cancel in sample and fail out of sample
- **Penalised regression**: minimise the squared error plus a penalty on the size of β
 - ▶ accept a little bias for a large reduction in variance (the trade-off of Section 2)

Ridge Regression (1/2)

- Hoerl & Kennard (1970): minimise the squared error plus a penalty on the squared coefficients

$$\hat{\beta}^{ridge} = \arg \min_{\beta} \sum_t (y_t - x_t^\top \beta)^2 + \lambda \sum_j \beta_j^2 = (X^\top X + \lambda I)^{-1} X^\top y$$

- Notation

- ▶ β_j : the coefficient of feature j ; $\sum_j \beta_j^2$: the penalty term, large when the coefficients are large
- ▶ $\lambda \geq 0$: the strength of the penalty; $\lambda = 0$ gives OLS, $\lambda \rightarrow \infty$ gives $\hat{\beta} \rightarrow 0$
- ▶ I : the $p \times p$ identity matrix

- Consequence

- ▶ $X^\top X + \lambda I$ is always invertible for $\lambda > 0$: ridge works even with more features than observations

Ridge Regression (2/2)

- ▣ **One feature**, centred (mean 0): $S_{xx} = \sum_t x_t^2$, $S_{xy} = \sum_t x_t y_t$

$$\hat{\beta}^{ridge} = \frac{S_{xy}}{S_{xx} + \lambda} = \frac{S_{xx}}{S_{xx} + \lambda} \hat{\beta}^{OLS}, \quad \hat{\beta}^{OLS} = \frac{S_{xy}}{S_{xx}}$$

- ▶ the factor $S_{xx}/(S_{xx} + \lambda)$ lies in $(0, 1]$: the share of the OLS coefficient that ridge keeps
- ▣ Example: $S_{xx} = 100$, $S_{xy} = 30$
 - ▶ OLS: $30/100 = 0.30$
 - ▶ ridge with $\lambda = 50$: $30/150 = 0.20$ (shrinkage factor 0.67)
- ▣ Ridge shrinks all coefficients towards zero, but sets none exactly to zero

The Lasso

- **Tibshirani (1996):** $\hat{\beta}^{lasso} = \arg \min_{\beta} \sum_t (y_t - x_t^\top \beta)^2 + \lambda \sum_j |\beta_j|$
 - ▶ the absolute value has a kink at 0: some coefficients become **exactly zero** (variable selection)
- **One feature: soft thresholding**
 - ▶ $\hat{\beta}^{lasso} = \text{sign}(S_{xy}) \max(|S_{xy}| - \lambda/2, 0) / S_{xx}$
 - ▶ $\text{sign}(S_{xy})$: +1 or -1; if $|S_{xy}| \leq \lambda/2$, the coefficient is exactly 0
 - ▶ same data, $\lambda = 20$: $(30 - 10)/100 = 0.20$; zero for $\lambda \geq 60$
- **In practice**
 - ▶ standardise the features first (the penalty depends on their units), with the training mean and standard deviation only
 - ▶ choose λ by walk-forward cross-validation; elastic net combines both penalties

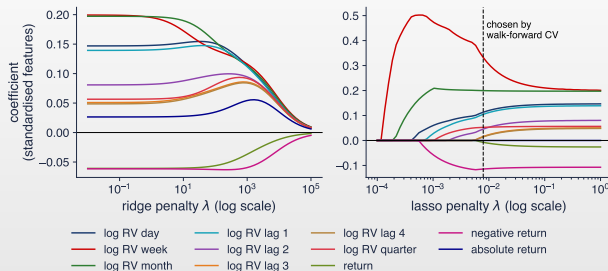


Robert Tibshirani, Stanford University

Photo: R. Tibshirani (2012); CC BY-SA 3.0;

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Ridge and Lasso Paths for the Volatility Features



- Target: log of the mean variance proxy of the next 5 days, S&P 500, 2008–2012 (1 193 days); 11 standardised features (Section 7)
- Lasso with $\lambda = 0.0079$ (walk-forward CV): 10 of 11 features kept; the week coefficient rises from +0.199 (OLS) to +0.283, the absolute return drops out

Interpretation of the Paths

- Ridge (left): all coefficients shrink smoothly and together; correlated features share the weight
 - ▶ the negative return keeps a negative sign until the end: falling markets raise future volatility (the leverage effect, [Chapter 9](#))
- Lasso (right): the features enter one by one as λ falls
 - ▶ the weekly variance enters first: the most informative single feature
 - ▶ at the chosen λ the lasso keeps the HAR structure (day, week, month) plus the negative return, -0.112
- The lasso selects; it does not test: a zero coefficient is not a p-value

Recap: Ridge and Lasso

- ▣ Penalties trade a little bias for less variance: $\lambda \sum \beta_j^2$ (ridge), $\lambda \sum |\beta_j|$ (lasso)
- ▣ Ridge shrinks, lasso shrinks and selects; standardise first, choose λ by walk-forward CV
- ▣ For volatility, the penalised model keeps the HAR structure and the leverage effect

Regression Trees

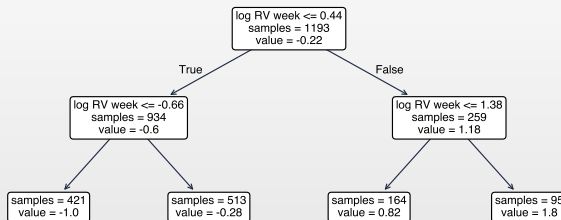
- A **regression tree** splits the feature space into rectangles and predicts the mean of y in each **leaf**
 - ▶ split rule: a feature j and a threshold s : left $x_j \leq s$, right $x_j > s$
- **Greedy fit**: at each node choose (j, s) that minimises $SSE = \sum_{\text{left}} (y - \bar{y}_L)^2 + \sum_{\text{right}} (y - \bar{y}_R)^2$
 - ▶ SSE: sum of squared errors; $\bar{y}_L$, $\bar{y}_R$: the means of y in the left and right groups (the two forecasts)
 - ▶ repeat in each half until a stopping rule: maximum depth, minimum number of observations per leaf
- Strengths and weaknesses
 - ▶ nonlinear effects and interactions without specifying them; insensitive to monotone transformations of x
 - ▶ high variance: a small change in the data can change the whole tree (Section 2)

Worked Example: One Split, Step by Step

s	$\bar{y}_L$	$\bar{y}_R$	SSE
0.70	0.70	1.52	1.708
0.90	0.80	1.68	1.248
1.25	0.80	1.97	0.227
1.75	1.00	2.15	0.505
2.20	1.22	2.20	1.468

- Six weeks: x = this week's volatility (%), y = next week's
 - ▶ x : 0.6, 0.8, 1.0, 1.5, 2.0, 2.4; y : 0.7, 0.9, 0.8, 1.6, 2.1, 2.2
 - ▶ no split: $\bar{y} = 1.38$, $\text{SSE} = 2.268$
- Candidates: the midpoints between consecutive x values
 - ▶ best split $s = 1.25$: leaves 0.80 and 1.97, $\text{SSE} = 0.227$
 - ▶ one split explains $1 - 0.227/2.268 = 0.90$ of the variation

A Depth-2 Tree for Next Week's Volatility



- S&P 500, 2008–2012, target and features in log variance ($\%^2$ per day); *value*: the mean of the target in the node
- Interpretation: the first split, $\log \text{RV week} \leq 0.44$, is an annual volatility of about 19.8%; all splits use the weekly variance, the feature the lasso picked first

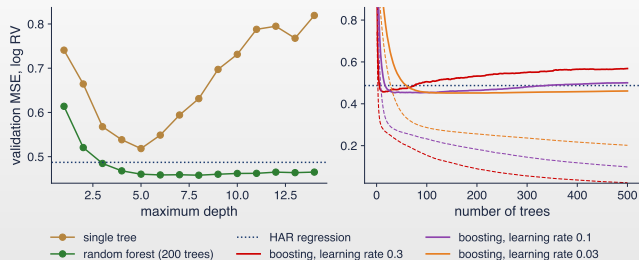
Random Forest

- Breiman (2001b): average B deep trees, each grown on a bootstrap sample (Chapter 2) of the training data (**bagging**)
 - ▶ at each split only a random subset of the features is tried: the trees become less correlated
- Why averaging works: B trees, each with variance σ^2 and pairwise correlation ρ
 - ▶ $\text{Var}(\text{average}) = \rho\sigma^2 + (1 - \rho)\sigma^2/B$: the part $\rho\sigma^2$, common to all trees, does not fall with B
 - ▶ with $\rho = 0.3$, $\sigma^2 = 1$: $B = 1$: 1.000; $B = 10$: 0.370; $B = 300$: 0.302
 - ▶ more trees never overfit; less correlated trees (fewer features per split) reduce the floor $\rho\sigma^2$
- Hyperparameters used here: 300 trees, at least 20 observations per leaf, half of the features tried at each split

Gradient Boosting

- **Friedman (2001)**: build the model step by step, each new small tree fitting what the previous ones missed
 - ▶ start with $F_0 = \bar{y}$; at step m , fit a shallow tree h_m to the residuals $y_t - F_{m-1}(x_t)$
 - ▶ update $F_m = F_{m-1} + \nu h_m$, with the **learning rate** $0 < \nu \leq 1$
 - ▶ for squared loss the residuals are minus the gradient of the loss: hence the name
- Random forest against boosting
 - ▶ forest: deep trees in parallel, averaging reduces variance
 - ▶ boosting: shallow trees in sequence, each step reduces bias; too many steps overfit
- Any differentiable loss works: quantile loss gives quantile boosting (Section 9)

One Tree, a Forest and Boosting on Volatility Data



- S&P 500, training 2008–2018 (2 698 days), validation 2019–2022 (1 008 days); dashed right: training MSE; dotted: HAR regression (0.487)
- Single tree: best 0.518 at depth 5, 0.819 at depth 14; forest: 0.458 (depth 8) and still 0.465 at depth 14

Interpretation: Averaging and Early Stopping

- The single tree overfits quickly; the forest is almost insensitive to depth
 - ▶ averaging removes most of the variance of deep trees; the forest beats HAR on this validation period
- Boosting: the validation error reaches its minimum, then rises while the training error keeps falling
 - ▶ learning rate 0.3: minimum 0.457 after 13 trees, 0.568 after 500
 - ▶ learning rate 0.03: minimum 0.451 after 154 trees, 0.461 after 500: smaller steps are safer
- The number of trees of boosting is a hyperparameter; the number of trees of a forest is not

Recap: Trees and Ensembles

- ▣ A tree splits greedily to minimise the SSE; alone it has high variance
- ▣ Random forest averages decorrelated deep trees; boosting adds shallow trees fitted to residuals
- ▣ Forest: robust to its hyperparameters; boosting: needs a small learning rate and a validated number of trees

From the Perceptron to the Multilayer Perceptron (1/2)

- A **neuron**: $h = g(b + w^\top x)$, a weighted sum of the inputs passed through an **activation function** g
 - ▶ x : the inputs; w : their weights; b : a constant (the bias term of the neuron); h : the output of the neuron
 - ▶ g : a fixed nonlinear function; without it, a network of neurons would be just a linear regression
- Activation functions
 - ▶ the perceptron uses a step function (0 or 1)
 - ▶ modern networks use smooth or piecewise linear g : logistic $1/(1 + e^{-z})$, tanh, ReLU $\max(0, z)$

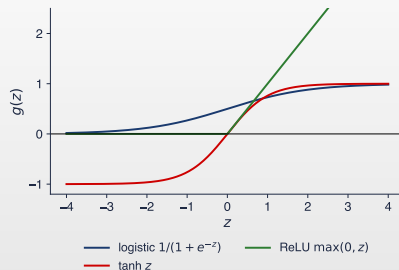
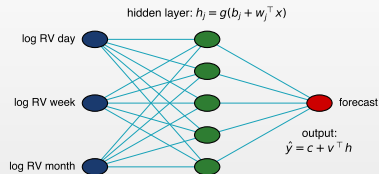
From the Perceptron to the Multilayer Perceptron (2/2)

- **MLP** (multilayer perceptron) with one hidden layer of K neurons

$$\hat{y} = c + \sum_{k=1}^K v_k g(b_k + w_k^\top x)$$

- ▶ w_k, b_k : the weights and the constant of hidden neuron k ; v_k : the weight of neuron k in the output; c : the output constant
- ▶ reading: a linear model in K nonlinear features $g(b_k + w_k^\top x)$, which are themselves learned
- Number of parameters with 11 inputs and 16 neurons
 - ▶ hidden layer: 11×16 weights + 16 constants = 192; output: $16 + 1 = 17$; total 209
- **Universal approximation** [Hornik, Stinchcombe & White \(1989\)](#): with enough neurons, one hidden layer approximates any continuous function on a compact set

A Small Network and Its Activation Functions



- Left: three HAR inputs, five hidden neurons, one forecast; each line is a weight
- Right: logistic and tanh saturate for large $|z|$; ReLU = $\max(0, z)$ is linear for $z > 0$ and trains faster

Training a Network

- Minimise the loss by **gradient descent**: $\theta \leftarrow \theta - \eta \nabla_{\theta} L$
 - ▶ θ : all weights and constants; $\nabla_{\theta} L$: the gradient, the vector of partial derivatives of the loss; $\eta > 0$: the learning rate (the step size)
 - ▶ each step moves θ in the direction in which the loss falls fastest
 - ▶ **back-propagation** computes $\nabla_{\theta} L$ layer by layer with the chain rule [Rumelhart, Hinton & Williams \(1986\)](#)
 - ▶ one **epoch** = one pass over the training data; Adam is a gradient method with adaptive step sizes
- The loss is not convex: different starting weights give different networks
 - ▶ we fix the random seeds and average five networks (an **ensemble**)
- Against overfitting
 - ▶ standardise inputs and target with training statistics; an L_2 penalty on the weights (as ridge); a fixed number of epochs
 - ▶ small data, small network: one hidden layer of 16 neurons is enough here

From Deep Blue to Deep Learning



IBM Deep Blue, Computer History Museum

Photo: Anton Chiang (2011); CC BY 2.0; Wikimedia Commons



Geoffrey Hinton, Nobel Prize in Physics 2024

Photo: Arthur Petron (2024); CC BY-SA 4.0; Wikimedia Commons

- 1997: Deep Blue beat the world chess champion mainly by search and hand-crafted evaluation rules, not by learning from data
- Deep learning (many layers, huge data) now dominates images and text; on financial prices the data are few and noisy, so small networks are the norm

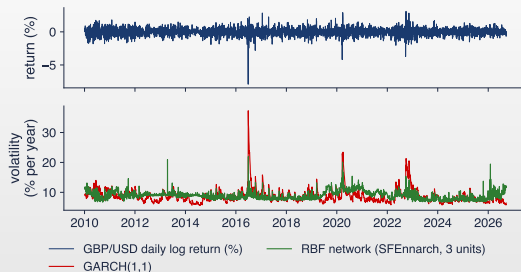
Neural-Network ARCH: the Textbook Model (1/2)

- Franke, Härdle & Hafner (2019), Ch. 19: $r_t = f(x_{t-1}) + s(x_{t-1})\varepsilon_t$, with f and s^2 estimated by neural networks
 - ▶ r_t : the return of day t ; x_{t-1} : the inputs known at the close of day $t - 1$
 - ▶ $f(x_{t-1})$: the conditional mean; $s^2(x_{t-1})$: the conditional variance; ε_t : a shock with mean 0 and variance 1
- ARCH (Engle (1982)) is the special case $s^2(x) = \omega + \sum_i \alpha_i r_{t-i}^2$
 - ▶ $\omega > 0$, $\alpha_i \geq 0$: the variance is a linear function of the past squared returns
 - ▶ the network imposes no shape on s^2

Neural-Network ARCH: the Textbook Model (2/2)

- The Quantlet **SFEnnarch**: GBP/USD, inputs x = the last 3 returns, the German 10-year yield, the gold return
 - ▶ step 1: a network $\hat{f}$ for the mean; residuals $e_t = r_t - \hat{f}(x_{t-1})$
 - ▶ step 2: a network for e_t^2 gives the conditional variance $\hat{s}^2(x_{t-1})$
- **RBF** (radial basis function) network with 3 units, as in the Quantlet
 - ▶ unit j : $\exp(-\|x - c_j\|^2/s_j^2)$; c_j : its centre (found by clustering); s_j : its width; $\|x - c_j\|$: the distance from x to c_j
 - ▶ the unit is close to 1 when x is near c_j and close to 0 far from it; output: a linear combination of the 3 units
 - ▶ nothing forces $\hat{s}^2 > 0$: a variance network needs a floor, which GARCH guarantees by its constraints

Neural-Network ARCH for GBP/USD



- Daily data from 7 January 2010 (4 048 days); annualised volatility of the RBF network and of GARCH(1,1) ($\hat{\alpha} = 0.077$, $\hat{\beta} = 0.892$)
- Average volatility: network 8.9%, GARCH 8.8%; correlation of the two paths 0.45 (a small MLP: 0.43)

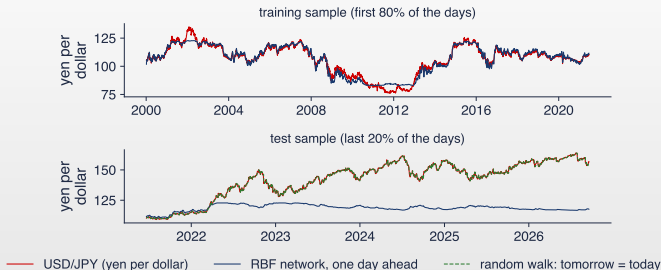
Interpretation: Flexible Shape, Short Memory

- After the Brexit vote, GARCH peaks at 37.3% (28 June 2016); the network only at 22.0%
 - ▶ the network sees only the last 3 returns: one large return raises its variance for 3 days, then the effect is gone
 - ▶ GARCH(1,1) (Chapter 9): $\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$, with σ_{t-1}^2 yesterday's conditional variance
 - ▶ the term $\beta \sigma_{t-1}^2$ carries the whole past: volatility clustering needs memory, not only flexibility
- The mean network explains 0.08% of the returns in sample: exchange rates are close to a random walk
 - ▶ the useful part of NN-ARCH is the variance, as for GARCH
- Lesson: good features (HAR averages, Section 7) matter more than the type of network

Forecasting an Exchange Rate Level: SFEnnjpgusd

- The Quantlet [SFEnnjpgusd](#) forecasts the yen-dollar rate from its last 3 values with an RBF network of 3 units
 - ▶ first 80% of the days for training, last 20% for testing; here USD/JPY from 2000, split on 21 June 2021
- Two traps of level forecasting
 - ▶ a level series looks predictable because $P_t \approx P_{t-1}$: the right benchmark is the random walk $\hat{P}_t = P_{t-1}$ ([Chapter 7](#))
 - ▶ a network with bounded units cannot extrapolate: the test levels go up to 163.9, the training maximum was 134.9
- The original code rescales the test forecasts with the range of the test series, which is unknown in real time; here the training range is used

The RBF Network against the Random Walk



- RMSE (root mean squared error) in yen: training: network 2.64, random walk 0.66; test: network 27.56, random walk 0.89
- Interpretation: out of sample the network is 31 times worse than "tomorrow = today"; on returns the same network has an out-of-sample R^2 of -0.14%

Recap: Neural Networks

- An MLP is a linear model in learned nonlinear features; trained by gradient descent and back-propagation
- NN-ARCH: a flexible variance function, but without the memory of GARCH and without positivity
- Forecast returns, not levels, and always compare with the random walk

Target and Features (1/2)

- Daily variance proxy of [Chapter 8](#) (no intraday data): an estimate of the variance of day t from its open, high, low and close

$$v_t = o_t^2 + \frac{1}{2}(u_t - d_t)^2 - (2 \ln 2 - 1)c_t^2$$

- ▶ o_t : the overnight log return (from the previous close to the open)
- ▶ u_t, d_t, c_t : log high, low and close relative to the open (the Garman–Klass term, [Garman & Klass \(1980\)](#))
- **Target:** $y_t = \ln \left(\frac{1}{5} \sum_{j=1}^5 v_{t+j} \right)$, the log realised variance of the next week
 - ▶ $j = 1, \dots, 5$: the next five trading days; the target is known only at the close of day $t + 5$
 - ▶ logs make the target closer to Normal and the forecasts positive after exp

Target and Features (2/2)

- ▣ **Features** at the close of day t
 - ▶ HAR: $\ln v_t$, $\ln$ of the mean over 5 days (week) and over 22 days (month)
 - ▶ extended (11): plus 4 lags of $\ln v_t$ ($\ln v_{t-1}, \dots, \ln v_{t-4}$), the 66-day mean, r_t , $\min(r_t, 0)$, $|r_t|$
- ▣ **Notation**
 - ▶ r_t : the log return of day t ; $|r_t|$: its absolute value
 - ▶ $\min(r_t, 0)$: the return if negative, 0 otherwise; it lets falls and rises act differently (the leverage effect)

The HAR Benchmark (1/2)

- **HAR** (heterogeneous autoregressive) model [Corsi \(2009\)](#)

$$y_t = \beta_0 + \beta_d \ln v_t + \beta_w \ln v_t^{(w)} + \beta_m \ln v_t^{(m)} + u_t$$

- ▶ $v_t^{(w)}$, $v_t^{(m)}$: the mean of v over the last 5 days (week) and over the last 22 days (month)
- ▶ β_0 : the constant; $\beta_d, \beta_w, \beta_m$: the weights of the three horizons; u_t : the error

- Interpretation

- ▶ three horizons of traders (daily, weekly, monthly); estimated by OLS
- ▶ with positive weights on the three horizons, a shock to volatility fades slowly
- ▶ a simple linear model that mimics long memory ([Chapter 11](#)) and is hard to beat

The HAR Benchmark (2/2)

- The question of this section
 - ▶ do lasso, random forest, gradient boosting or an MLP add anything to HAR out of sample?
 - ▶ a large study with intraday data finds modest gains, mainly from extra features [Christensen, Siggaard & Veliyev \(2023\)](#)
- Design: walk-forward of Section 3, yearly refits; S&P 500 from 2013, Bitcoin from 2018
 - ▶ variance forecast $\hat{h}_t = \exp(\hat{y}_t + \hat{s}^2/2)$, $\hat{s}^2$ the residual variance in training (the log-Normal mean)
 - ▶ $\hat{y}_t$: the forecast of the log variance; $\exp(\hat{y}_t)$ alone would underestimate the mean variance, hence the term $\hat{s}^2/2$

Evaluation (1/3): Out-of-Sample R^2

- **Out-of-sample R^2** against a benchmark forecast $\tilde{y}_t$ [Campbell & Thompson \(2008\)](#)

$$R_{OOS}^2 = 1 - \frac{\sum_t (y_t - \hat{y}_t)^2}{\sum_t (y_t - \tilde{y}_t)^2}$$

- ▶ $\hat{y}_t$: the forecast of the model; sums over the test days only
- ▶ benchmark $\tilde{y}_t$: the training mean, or HAR
- Reading
 - ▶ $R_{OOS}^2 \leq 1$; $R_{OOS}^2 > 0$: smaller squared error than the benchmark; $R_{OOS}^2 < 0$: worse than the benchmark
 - ▶ unlike the in-sample R^2 , it can be negative
- Step by step, S&P 500, MLP against HAR
 - ▶ $1 - 0.397/0.433 = 1 - 0.917$, i.e. +8.3%

Evaluation (2/3): QLIKE

- ▣ **QLIKE** (Chapter 8): a loss for variance forecasts, robust to a noisy proxy Patton (2011)

$$L_t = \frac{RV_t}{\hat{h}_t} + \ln \hat{h}_t$$

- ▶ RV_t : the realised variance of the next 5 days (the target, in levels); $\hat{h}_t$: its forecast
- ▶ for a given RV_t , the loss is smallest at $\hat{h}_t = RV_t$; lower average QLIKE is better
- ▣ Asymmetry
 - ▶ for $RV = 1$: $\hat{h} = 0.5$ costs 1.307, $\hat{h} = 2$ costs 1.193
 - ▶ under-prediction is punished more: for risk management, underestimating volatility is the costly error

Evaluation (3/3): the Diebold–Mariano Test

- **Diebold–Mariano** test [Diebold & Mariano \(1995\)](#): do two models A and B have the same expected loss?

$$d_t = L_t^A - L_t^B, \quad \text{DM} = \frac{\bar{d}}{\sqrt{\widehat{\text{LRV}}/n}} \sim N(0, 1) \text{ under equal accuracy}$$

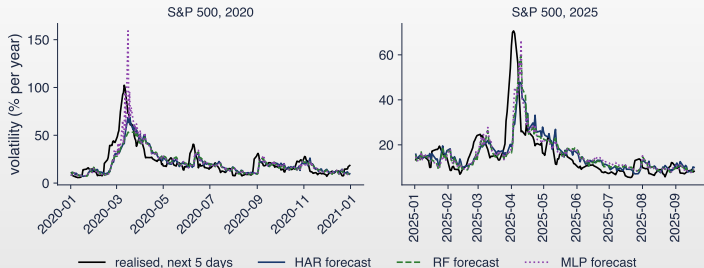
- ▶ L_t^A, L_t^B : the losses of the two models on day t ; $\bar{d}$: the mean of d_t ; n : the number of test days
- ▶ $\widehat{\text{LRV}}$: the Newey–West long-run variance of d_t (5 lags), because overlapping targets make d_t autocorrelated
- Reading
 - ▶ $\text{DM} < 0$: A has the smaller loss; $\text{DM} > 0$: B is better
 - ▶ $|\text{DM}| > 1.96$: the difference is significant at 5%

Results: S&P 500 and Bitcoin

	Model	R^2_{OOS} vs. mean	R^2_{OOS} vs. HAR	QLIKE	DM vs. HAR (p)
S&P 500	HAR	61.5	–	0.414	–
	Lasso	64.2	+7.1	0.400	–5.12 (< 0.001)
	RF	61.9	+1.0	0.416	+0.32 (0.747)
	GB	60.4	–2.7	0.421	+1.02 (0.308)
	MLP	64.7	+8.3	0.398	–3.96 (< 0.001)
Bitcoin	HAR	44.0	–	3.111	–
	Lasso	45.1	+2.0	3.108	–0.64 (0.520)
	RF	43.4	–1.2	3.119	+0.81 (0.418)
	GB	41.7	–4.2	3.148	+2.12 (0.034)
	MLP	43.8	–0.3	3.123	+1.10 (0.272)

- Forecasts for 3 444 days (S&P 500, from 2 January 2013) and 3 178 days (Bitcoin, from 1 January 2018); R^2_{OOS} in %, on log RV; GB: gradient boosting
- S&P 500: lasso and MLP beat HAR significantly (+7.1%, +8.3%); random forest and boosting do not
- Bitcoin: no model beats HAR; boosting is significantly worse (DM +2.12)

Forecasts in Two Storms: 2020 and 2025

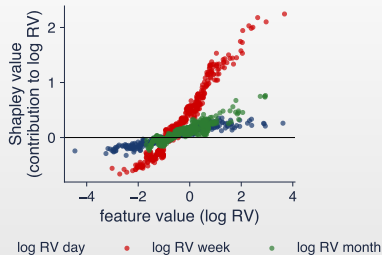
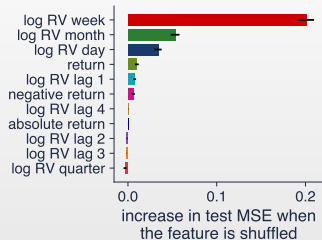


- S&P 500: annualised realised volatility of the next 5 days, $\sqrt{252 RV_t}$, and the HAR, random forest and MLP forecasts made at the close of day t
- Interpretation: all models react only after the first shock (March 2020, April 2025); the forest is capped by the largest values it saw in training, the MLP can overshoot

Feature Importance: Permutation and Shapley Values

- ▣ **Permutation importance:** shuffle one feature in the test data and measure how much the test MSE rises
 - ▶ large increase: the model relies on the feature; correlated features share (and hide) their importance
- ▣ **Shapley values** (game theory): $\hat{f}(x) = \phi_0 + \sum_j \phi_j(x)$, ϕ_j the fair share of feature j in the forecast
 - ▶ ϕ_0 : the average forecast; $\phi_j(x) > 0$: feature j raises this forecast above the average, $\phi_j(x) < 0$: lowers it
 - ▶ ϕ_j averages the change of $\hat{f}$ when j joins every subset of the other features
 - ▶ SHAP (Shapley additive explanations) [Lundberg & Lee \(2017\)](#): fast versions for trees; for a linear model $\phi_j = \beta_j(x_j - \bar{x}_j)$
- ▣ Both describe the model, not the economy: importance is not causality

The Features Used by the Forest



- Left: random forest on 11 features, trained to 2019, tested 2020–2026 (1 682 days): shuffling the weekly variance raises the MSE by 0.201, the monthly by 0.053, the daily by 0.034
- Right: exact Shapley values of a forest on the 3 HAR features; mean $|\phi|$: week 0.52, month 0.13, day 0.13; the effect is nonlinear and grows with the level of volatility

Recap: Forecasting Realised Volatility

- HAR is a strong benchmark; penalised and neural models add a few per cent on the S&P 500, nothing on Bitcoin
- Evaluate with out-of-sample R^2 , QLIKE and Diebold–Mariano, on a walk-forward design
- The weekly variance carries most of the information; importance measures describe the model, not causes

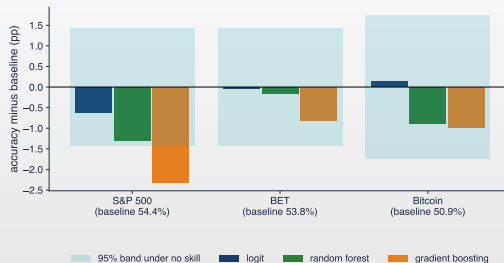
A Classification Problem

- Target: $y_t = 1\{r_{t+1} > 0\}$; features at the close of day t : the last 5 returns, sums over 5, 21 and 63 days, volatility over 21 and 63 days
 - ▶ $1\{\cdot\}$: the indicator, 1 if the condition holds and 0 otherwise; $y_t = 1$: tomorrow's return is positive
 - ▶ models: logit ([Chapter 12](#)), random forest and gradient boosting classifiers; walk-forward, yearly refits
 - ▶ S&P 500 and BET from 2008, Bitcoin from 2018
- Metrics (defined in [Chapter 12](#))
 - ▶ accuracy: the share of days with the right sign; AUC (area under the ROC curve): 0.5 means no ranking skill
- Why expect little: weak-form efficiency ([Chapter 7](#))
 - ▶ any reliable pattern in past returns would be traded away

The Right Baseline

- ▣ **Majority-class baseline:** always forecast the more frequent class of the training sample (here: “up”)
 - ▶ S&P 500 since 2008: 54.4% of the days are up, so “always up” is right on 54.4% of the days
 - ▶ an accuracy of 53% sounds like skill; against this baseline it is a loss
- ▣ Test against the baseline: $z = (\widehat{acc} - acc_0) / \sqrt{acc_0(1 - acc_0)/n}$
 - ▶ $\widehat{acc}$: the accuracy of the model; acc_0 : that of the baseline; n : the test days; $z \approx N(0, 1)$ if the model has no extra skill
 - ▶ S&P 500, $n = 4705$: standard error 0.73 points; a 95% band of ± 1.4 points around the baseline
 - ▶ random forest: 53.1%, $z = -1.79$ (p 0.074); boosting: 52.1%, $z = -3.19$ (p 0.001)
- ▣ For returns, the matching baseline is the zero or historical-mean forecast [Welch & Goyal \(2008\)](#)

Accuracy against the Baseline



- Accuracy minus the baseline accuracy (percentage points); band: ± 1.96 standard errors around zero
- AUC of the logit and of the forest: S&P 500 0.511 and 0.507, BET 0.533 and 0.533, Bitcoin 0.515 and 0.519; logit forecasts “up” on 86% of S&P 500 days

Interpretation: Why Sign Prediction Mostly Fails

- No model beats “always up” significantly on any market; the flexible models do worse than the logit
 - ▶ S&P 500: logit -0.6 , forest -1.3 , boosting -2.3 points; Bitcoin: logit $+0.1$, forest -0.9
- The signal-to-noise ratio of daily returns is tiny
 - ▶ flexible models find patterns in the noise of the training years (Section 2); the logit is close to the baseline because it hardly moves
- Even a real edge of one point would rarely survive costs: a daily sign strategy trades very often
 - ▶ ML gains in returns come from large cross-sections at monthly horizons, not from one index tomorrow (Section 11)

Recap: The Sign of Tomorrow's Return

- Always compare accuracy with the majority-class baseline, and test the difference
- On the S&P 500, BET and Bitcoin no model beats “always up” significantly
- Weak-form efficiency ([Chapter 7](#)) predicts this result

Quantile Regression (1/2)

- Koenker & Bassett (1978): estimate the conditional α -quantile $q_\alpha(x) = x^\top \beta_\alpha$ by minimising the **quantile (pinball) loss**
 - ▶ $q_\alpha(x)$: the level below which y falls with probability α , given x ; β_α : coefficients specific to the level α
- The loss of an error $u = y - q$:

$$\rho_\alpha(u) = u(\alpha - 1\{u < 0\})$$

- ▶ below the quantile ($u < 0$) the error costs $(1 - \alpha)|u|$, above it $\alpha|u|$
- ▶ with $\alpha = 0.01$, the minimum is reached when about 1% of the observations lie below q

Quantile Regression (2/2)

- ▣ With $\alpha = 0.01$ and $q = -\text{VaR}$: the average quantile loss of [Chapter 10](#)
 - ▶ $(\alpha - I_t)(r_t + \text{VaR}_t)$, with $I_t = 1$ if $r_t < -\text{VaR}_t$ (an exception) and 0 otherwise
- ▣ VaR 1% for day $t + 1$: $\widehat{\text{VaR}}_t = -\hat{q}_{0.01}(x_t)$, with x_t = daily, weekly and monthly volatility (square roots of the proxy) and r_t
 - ▶ no distribution is assumed: the data choose how volatility maps into the tail; CAViaR [Engle & Manganelli \(2004\)](#) is a dynamic version
- ▣ **Quantile gradient boosting**: the same loss with boosted trees (Section 5)
 - ▶ flexible, but at 1% each leaf sees few tail observations: high variance

Backtest of VaR 1%: Quantile Methods, HS and GARCH-t (1/2)

	Method	x	rate (%)	p Kupiec	p ind.	loss $\times 100$	mean VaR
S&P 500	QR	28	0.81	0.252	0.020	3.38	2.70
	QGB	42	1.22	0.213	0.015	3.51	2.46
	HS	46	1.33	0.061	< 0.001	4.78	2.97
	GARCH-t	52	1.51	0.005	0.008	3.50	2.42
Bitcoin	QR	23	0.72	0.098	0.563	13.03	10.46
	QGB	47	1.48	0.012	0.235	13.19	8.86
	HS	32	1.01	0.974	0.332	13.01	9.30
	GARCH-t	41	1.29	0.117	0.558	13.15	8.52

Columns

- ▶ x : the number of exceptions ($r_{t+1} < -\widehat{\text{VaR}}_t$) in n test days; rate = x/n , target 1%
- ▶ p Kupiec: p-value of the Kupiec test of the rate; p ind.: p-value of the Christoffersen test of independence ([Chapter 10](#))
- ▶ loss $\times 100$: the mean quantile loss (smaller is better); mean VaR: the average VaR 1%, in %

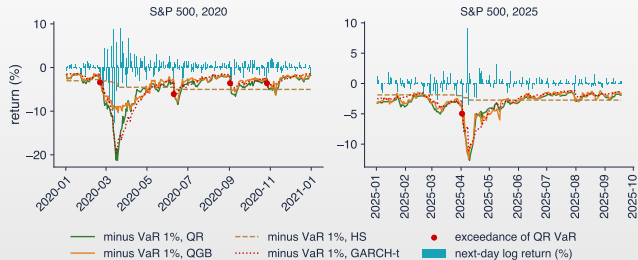
Methods

- ▶ QR: linear quantile regression; QGB: quantile gradient boosting
- ▶ HS: historical simulation on 500 days; GARCH-t: re-estimated each January ([Chapter 10](#))

Backtest of VaR 1%: Quantile Methods, HS and GARCH-t (2/2)

- S&P 500 (3 448 days from 2 January 2013): QR has the right rate (0.81%) and the smallest loss
 - ▶ its exceptions are not fully independent (p ind. 0.020)
 - ▶ GARCH-t is exceeded too often (1.51%)
- Bitcoin (3 182 days): HS has the best rate (1.01%)
 - ▶ QGB is rejected by the Kupiec test (1.48%, p 0.012)
 - ▶ at 1% each leaf of the boosted trees sees few tail days: high variance (Section 2)
- Reading the p-values
 - ▶ p Kupiec below 0.05: the exception rate differs significantly from 1%
 - ▶ p ind. below 0.05: the exceptions come in clusters, so the VaR reacts too slowly

VaR 1% Forecasts in 2020 and 2025



- S&P 500: next-day returns and minus VaR 1% of the four methods; dots: exceedances of the quantile-regression VaR
- Interpretation: QR and GARCH-t move with volatility; HS reacts only slowly; in 2020 HS was exceeded 10 times, QR 4 times, and the HS exceptions cluster (p ind. < 0.001)

Recap: VaR 1% by Quantile Methods

- ▣ Quantile regression minimises the pinball loss: VaR without a distributional assumption
- ▣ A linear quantile model on volatility features passes the Kupiec test for the S&P 500
- ▣ Flexible quantile models at 1% need much more tail data than we have

The Best of N Strategies (1/2)

- ▣ **Data snooping:** trying many models or rules on the same data and reporting the best White (2000)
 - ▶ with enough trials, a rule without skill looks excellent in sample
 - ▶ hundreds of published return predictors would not survive a multiple-testing correction Harvey, Liu & Zhu (2016)
- ▣ The annual Sharpe ratio of a strategy without skill over Y years is about $N(0, 1/Y)$
 - ▶ Sharpe ratio: mean excess return divided by its standard deviation (Chapter 1); Y : the number of years of data
 - ▶ mean 0: no skill; variance $1/Y$: the estimation error, which falls with the length of the sample

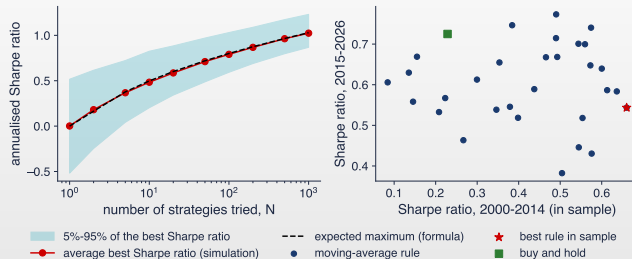
The Best of N Strategies (2/2)

- Expected maximum of N independent zero-skill trials [Bailey & López de Prado \(2014\)](#)

$$E[\max] \approx \sigma \left[(1 - \gamma) \Phi^{-1} \left(1 - \frac{1}{N} \right) + \gamma \Phi^{-1} \left(1 - \frac{1}{Ne} \right) \right]$$

- ▶ σ : the standard deviation of the Sharpe ratios across trials ($1/\sqrt{Y}$ above); N : the number of trials
- ▶ Φ^{-1} : the quantile function of the standard Normal distribution; $e \approx 2.718$; $\gamma = 0.5772$: the Euler–Mascheroni constant
- Example: 10 years, $\sigma = 1/\sqrt{10} = 0.316$, $N = 100$
 - ▶ $0.316 [(1 - \gamma) 2.326 + \gamma 2.680] = 0.80$: a Sharpe ratio this large is expected from pure luck
 - ▶ simulation: the best of 10, 100 and 1000 zero-skill strategies has an average Sharpe ratio of 0.48, 0.79, 1.02

Selection Bias in Practice



- Left: the best annual Sharpe ratio among N strategies without skill (10 years); right: 30 moving-average rules on the S&P 500 (long or cash), 2000–2014 against 2015–2026
- Interpretation: the best rule in sample (MA 50/200, Sharpe 0.66) earns 0.54 afterwards, rank 22 of 30, below buy and hold (0.72); in and out of sample Sharpe ratios correlate 0.04

The Deflated Sharpe Ratio (1/2)

- Bailey & López de Prado (2014): the probability that the true Sharpe ratio exceeds what the best of N trials would show by luck

$$\text{DSR} = \Phi\left(\frac{(\widehat{SR} - SR_0)\sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3\widehat{SR} + \frac{\hat{\gamma}_4-1}{4}\widehat{SR}^2}}\right)$$

- Notation

- ▶ $\widehat{SR}$: the per-day Sharpe ratio of the chosen strategy; T : the number of days
- ▶ SR_0 : the expected maximum of N zero-skill trials (above), with σ = the standard deviation of the N Sharpe ratios tried
- ▶ $\hat{\gamma}_3$: the skewness, $\hat{\gamma}_4$: the kurtosis of the daily returns (3 for the Normal distribution); Φ : the standard Normal distribution function

- Reading

- ▶ the DSR is a probability in $(0, 1)$; above 0.95: the Sharpe ratio is significant after the correction for N trials
- ▶ negative skewness and fat tails enlarge the denominator and lower the DSR

The Deflated Sharpe Ratio (2/2)

- ▣ **Step by step:** the best MA rule, $N = 30$, $T = 3\,769$ days
 - ▶ $\widehat{SR} = 0.0416$ per day ($\times \sqrt{252} = 0.66$); $\hat{\gamma}_3 = -0.44$, $\hat{\gamma}_4 = 12.0$
 - ▶ $SR_0 = 0.0105 [(1 - \gamma) 1.834 + \gamma 2.249] = 0.0219$ (annual 0.35)
 - ▶ z , the argument of Φ in the DSR formula: $z = 1.20$, so $DSR = \Phi(z) = 0.88$
 - ▶ without the correction for N (PSR, probabilistic Sharpe ratio, $SR_0 = 0$): 0.994
- ▣ Interpretation: alone the rule looks significant; after 30 trials it is not (DSR below 0.95), as its later performance confirms

Recap: Data Snooping

- ▣ The best of many trials is biased upwards; the bias grows like $\sqrt{2 \ln N}$
- ▣ Report the number of trials and deflate the Sharpe ratio
- ▣ A final test sample used only once is the simplest protection

Empirical Asset Pricing via Machine Learning

- **Gu, Kelly & Xiu (2020):** which methods best predict the monthly return of individual US stocks?
 - ▶ nearly 30,000 stocks, 1957–2016; 94 firm characteristics, interacted with 8 macroeconomic variables, plus 74 industry dummies: 920 features
- **Design:** 18 years of training (1957–1974), 12 of validation (1975–1986), 30 of test (1987–2016)
 - ▶ refit once a year; the training window grows, the validation window rolls forward: a walk-forward design
 - ▶ models: OLS, penalised and dimension-reduction regressions, random forest, boosted trees
 - ▶ networks NN1–NN5: NN k has k hidden layers

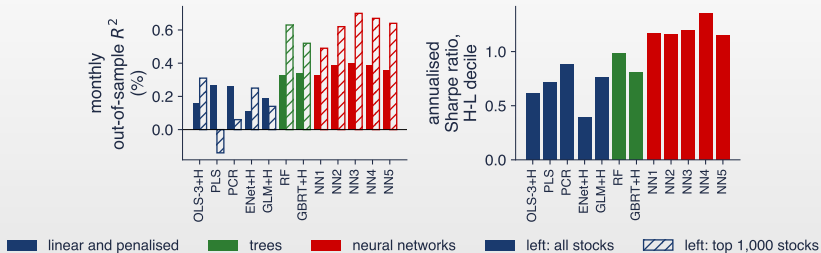


Charles M. Harper Center, Chicago Booth, where
Gu and Xiu worked

Photo: Michael Barera (2013); CC BY-SA 4.0;
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Evaluation in the Paper

- $R_{OOS}^2 = 1 - \sum_{(i,t)} (r_{i,t+1} - \hat{r}_{i,t+1})^2 / \sum_{(i,t)} r_{i,t+1}^2$: the benchmark is a forecast of **zero**, not the historical mean
 - ▶ $r_{i,t+1}$: the return of stock i in month $t+1$; $\hat{r}_{i,t+1}$: its forecast; the sums run over all stock-months of the test period
 - ▶ the authors note that the historical mean of a stock is so noisy that it is easy to beat: a zero benchmark is harder
- Diebold–Mariano tests between all pairs of models, with a Bonferroni correction for the 12 comparisons
 - ▶ the same tool as Section 7, applied to a panel of stocks
- Economic value: each month, sort stocks into deciles by forecast; buy the top decile, sell the bottom one
 - ▶ value-weighted portfolios; Sharpe ratio, drawdown and turnover (the share of the portfolio traded each month)

Results: Small R^2 , Large Economic Value

- Left: monthly R^2_{OOS} (Table 1): OLS with all 920 features -3.46% ; OLS-3 (size, value, momentum) 0.16% ; random forest 0.33% ; NN3 0.40% (0.70% for the 1,000 largest stocks)
- Right: annual Sharpe ratio of the value-weighted decile spread (Table 7): OLS-3 0.61 , random forest 0.98 , NN4 1.35

Interpretation and Limits

- ▣ What the paper shows
 - ▶ nonlinear models (trees, networks) beat linear ones because they capture interactions between characteristics
 - ▶ all methods agree on the main signals: price trends (momentum, short-term reversal), liquidity, volatility
 - ▶ shallow networks are enough: 3–4 layers beat 5; “deep” learning does not help with monthly returns
- ▣ What to keep in mind
 - ▶ an R^2_{OOS} below 1% per month is a large result here: compare with the sign forecasts of Section 8
 - ▶ neural portfolios turn over more than 100% of their value each month: trading costs reduce the gains
 - ▶ the evidence is a cross-section of thousands of stocks; it says little about timing one index

An Open Question

- **Do machine-learning volatility forecasts help on the Bucharest Stock Exchange, where liquidity is low and intraday data are scarce?**
 - ▶ today: on the S&P 500 the lasso gains +7.1% over HAR; on Bitcoin no significant gain (+2.0%, p 0.520)
 - ▶ the BET has no high and low prices in our data, so the proxy must come from squared returns, which are noisier
- Why it is open: thin trading, a different calendar from New York, few crises in the sample
- Related work, as further reading: [Pele et al. \(2023\)](#) (statistical classification of crypto assets); [Pele & Mazurencu-Marinescu-Pele \(2026\)](#) (how precise forecast comparisons can be)
- AI tools can speed up such a study; they do not replace checking it [Wang et al. \(2023\)](#)

How AI Could Help

- ▣ **Literature:** a list of studies on HAR and machine learning for emerging and frontier markets
- ▣ **Code:** a first draft of the walk-forward comparison of HAR, lasso and random forest for the BET and three BVB stocks
- ▣ **Robustness:** other proxies (absolute returns, two-day returns), other horizons (1, 5, 22 days), S&P 500 features of the previous close
- ▣ Example prompt
 - ▶ Write Python code that compares HAR, lasso and a random forest for the weekly log variance of the BET index (proxy: squared daily returns), with yearly walk-forward refits from 2012, purging the last 5 training rows, and reports the out-of-sample R2, QLIKE and Diebold-Mariano tests.

What to Check

- ▣ Time order: no shuffled folds; features use only data up to the close of day t ; the last training targets do not reach into the test period
- ▣ Scaling and tuning inside the training window only (standardisation, λ , the number of trees)
- ▣ Calendars: the BVB closes before New York; S&P 500 features must come from the previous New York close
- ▣ A benchmark in every table (HAR, the majority class, historical simulation) and a test of the difference
- ▣ The number of models tried, reported; references: every cited paper must exist; check the DOI

Project Idea

- **Question:** do machine-learning models forecast the volatility of BVB stocks better than HAR?
 - ▶ data: BET, Banca Transilvania, OMV Petrom, BRD (EODHD); S&P 500 and DAX as external features
- Steps
 - ▶ a variance proxy for each stock (range where available, squared returns otherwise), weekly target
 - ▶ HAR, lasso, random forest and a small MLP, walk-forward from 2014, yearly refits with purging
 - ▶ out-of-sample R^2 , QLIKE and Diebold–Mariano; Shapley values for the external features
- Deliverable: one table, one chart, and a paragraph on what the data can and cannot show
- Declare any AI use, and list the errors of the AI that you corrected

Key Takeaways

- ▣ Machine learning is judged by its error on new data; the bias–variance trade-off sets how flexible a model should be
- ▣ Time series need walk-forward validation with purging; random folds on overlapping targets fake predictability
- ▣ Ridge and lasso shrink; forests average trees; boosting adds trees; networks learn nonlinear features
- ▣ Volatility: HAR is hard to beat, the gains are a few per cent; the sign of returns: no gain over “always up”
- ▣ VaR 1%: linear quantile regression works; flexible quantile models lack tail data
- ▣ Report the number of trials and deflate the Sharpe ratio; compare every model with a simple benchmark

Key Formulas

Quantity	Formula
Bias–variance	$E[(y_0 - \hat{f})^2] = (E\hat{f} - f)^2 + \text{Var}(\hat{f}) + \sigma^2$
Ridge	$(X^\top X + \lambda I)^{-1} X^\top y$; 1D: $S_{xy} / (S_{xx} + \lambda)$
Lasso (1D)	$\text{sign}(S_{xy}) \max(S_{xy} - \lambda/2, 0) / S_{xx}$
Random forest	$\text{Var} = \rho\sigma^2 + (1 - \rho)\sigma^2/B$
Boosting	$F_m = F_{m-1} + \nu h_m$
MLP	$\hat{y} = c + \sum_k v_k g(b_k + w_k^\top x)$
HAR	$y_t = \beta_0 + \beta_d \ln v_t + \beta_w \ln v_t^{(w)} + \beta_m \ln v_t^{(m)} + u_t$
R_{OOS}^2	$1 - \sum (y - \hat{y})^2 / \sum (y - \tilde{y})^2$
Quantile loss	$\rho_\alpha(u) = u(\alpha - 1\{u < 0\})$
DSR	$\Phi((\widehat{SR} - SR_0)\sqrt{T-1}/\sqrt{1 - \hat{\gamma}_3\widehat{SR} + (\hat{\gamma}_4 - 1)\widehat{SR}^2/4})$

Check Yourself

- **Question:** a model has a training R^2 of 80% and an out-of-sample R^2 of -5% . What happened?
 - ▶ **Answer:** overfitting: the model learned the noise of the training sample and loses against the benchmark on new data
- **Question:** why is random K-fold wrong for a 21-day return target?
 - ▶ **Answer:** neighbouring targets overlap, so the test answers leak into training; use walk-forward with purging
- **Question:** a classifier is right on 53% of the days and the market rose on 54% of them. Is it useful?
 - ▶ **Answer:** no: “always up” is right on 54% of the days
- Next: [Chapter 14](#), crypto assets

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